

CMAN :=

Given data $N = 32$, $FP32 = 4$ bytes , $DRAM \text{ bandwidth} = 320 \text{ GB/s}$
Compute = 10 TFLOPS

Task 1:

$N = 32$ times , So each $B[K][J]$ accessed 32 times

- A accesses : $N^3 = 32^3 = 32,768$ elements
- B accesses : $N^3 = 32^3 = 32,768$ elements

Total elements accessed :

$$2N^3 = 2 \times 32^3 = 65,536$$

Total DRAM bytes :

$$65,536 \times 4 = 262,144 \text{ bytes} = 256 \text{ kB}$$

Total DRAM traffic (naive) = 256 kB .

Task 2:

Tile size $T = 8$

Number of tiles per dimension: $N / T = 32 / 8 = 4$

Each A and B tile is reused $N/T = 4$ times

A tile loads:

$$(N/T)^3 \times T^2 = (4)^3 \times 8^2 = 64 \times 64 = 4096 \text{ elements}$$

B tile loads:

$$4096 \text{ elements} .$$

Total DRAM traffic :

$$2 \times 4096 \times 4 = 32,768 \text{ bytes} = 32 \text{ kB}$$

total DRAM traffic (tilled) : 32 kB .

Task 3:

Naive traffic = 262,144 bytes

Tiled traffic = 32,768 bytes

Traffic ratio : Naive / Tiled = $262,144 / 32,768 = 8$ T

If we consider only B's reuse :

Tiled traffic = $262,144 / 32 = 8,192$ bytes

Ratio : $262,144 / 8,192 = 32 = N$

Explanation:

In the naive matrix multiplication, matrix B is entirely re-read from DRAM for each row of matrix A, since every output element requires a full traversal of B. Consequently, matrix B is reloaded N times across the entire computation. In the tiled approach, each $T \times T$ sub-block of B is loaded from DRAM exactly once and retained in on-chip shared memory for reuse within that tile. Therefore, naive DRAM traffic is exactly N times larger than tiled DRAM traffic.

Task 4:

Bandwidth = 320 GB/s

Compute = 10 TFLOPS

Total FLOPs = $2 \times N^3 = 2 \times 32^3 = 65,536$ FLOPs

Naive Case:

Memory time: $t_{\text{memory}} = \text{Traffic} / \text{BW}$

$$= 262,144 / (320 \times 10^9)$$

$$= 8.192 \times 10^{-7} \text{ s} = 0.819 \text{ ms}$$

Compute time:

$$t_{\text{compute}} = \text{FLOPs} / P$$

$$= 65,536 / 10^{13}$$

$$= 6.554 \times 10^{-9} \text{ s} = 0.00655 \text{ ms}$$

$$t_{\text{memory}} (0.819\text{ms}) \gg t_{\text{compute}} (0.00655\text{ms})$$

Execution time (naive) : 0.819ms

Bottleneck: Memory-bound.

Tiled Case (T=8)

Memory time:

$$t_{\text{memory}} = 32,768 / (320 \times 10^9)$$

$$= 1.024 \times 10^{-7} \text{ s} = 0.1024\text{ms}$$

$$\text{Compute time: } \text{FLOPs} / P = 65,536 / (1.0 \times 10^{13})$$

$$\Rightarrow 6.554 \times 10^{-9} \text{ s} = 0.00655\text{ms}$$

$$t_{\text{memory}} (0.1024 \text{ ms}) > t_{\text{compute}} (0.00655\text{ms})$$

Execution time(tiled) = 0.1024ms

Bottleneck : Memory-bound.